Bridge sampling schemes, and what separates them

normalised time $s = t/T \in [0,1]$

The process. Source $x \sim \Phi = \mathcal{N}(0, I)$, target $y \sim \Psi$, driver $Z_t = x + \sigma W_t$. The Φ-initialised Gaussian random bridge to Ψ is $\xi_t \mid (x,y) \sim \mathcal{N}\big(x + \frac{t}{T}(y-x),\ \sigma^2 \frac{t(T-t)}{T} I\big)$, i.e.

$$\xi_s = (1-s)x + sy + \gamma\sqrt{s(1-s)}\,\varepsilon, \qquad \varepsilon \sim \mathcal{N}(0,I), \quad \gamma^2 := \sigma^2 T. \tag{1}$$

σ and T are not separately identifiable — training and sampling depend on them only through γ; the published $\sigma = 1, T = 0.1$ is $\gamma = 0.3162$. The network regresses the endpoint, $f_\theta(\xi_s, s) \approx \mathbb{E}[Y \mid \xi_s]$, under $\mathbb{E}\|y - f_\theta\|^2$ with $s \sim \mathcal{U}[0,1)$ — not a score, not a noise prediction. The non-anticipative dynamics are

$$\mathrm{d}\xi_s = \frac{\mathbb{E}[Y \mid \xi_s] - \xi_s}{1-s}\,\mathrm{d}s + \gamma\,\mathrm{d}B_s. \tag{2}$$

Three samplers. With $\hat{y} = f_\theta(\xi_s, s)$, advancing $s \to u$; each costs one network evaluation.

(a) Euler–Maruyama — Algorithm 2 of the manuscript, a discretisation of (2): $\xi_u = \xi_s + \frac{\hat{y}-\xi_s}{1-s}(u-s) + \gamma\sqrt{u-s}\,z$, $z \sim \mathcal{N}(0,I)$.

(b) Deterministic Euler — (2) with the diffusion deleted; **this is what the original code evaluated** (`sample_bridges.py` passes `deterministic=True`): $\xi_u = \xi_s + \frac{\hat{y}-\xi_s}{1-s}(u-s)$. It is *biased*. A deterministic sampler preserving the marginals of (2) must follow the probability-flow ODE, with drift $b(\xi,s) - \frac{\gamma^2}{2}\nabla \log p_s(\xi)$; dropping $\gamma\,\mathrm{d}B_s$ without that score correction gives neither the SDE nor its flow ODE, so refining the grid converges to the wrong law. The symptom is under-dispersion, and it is the reported step-count plateau.

(c) Exact bridge posterior — recommended. Given ξ_s and $\hat{y}$, the law of ξ_u under the Brownian bridge pinned at (s, ξ_s) and $(1, \hat{y})$ is exactly

$$\xi_u \sim \mathcal{N}\Big(\xi_s + \tfrac{u-s}{1-s}(\hat{y} - \xi_s),\ \gamma^2 \tfrac{(u-s)(1-u)}{1-s} I\Big). \tag{3}$$

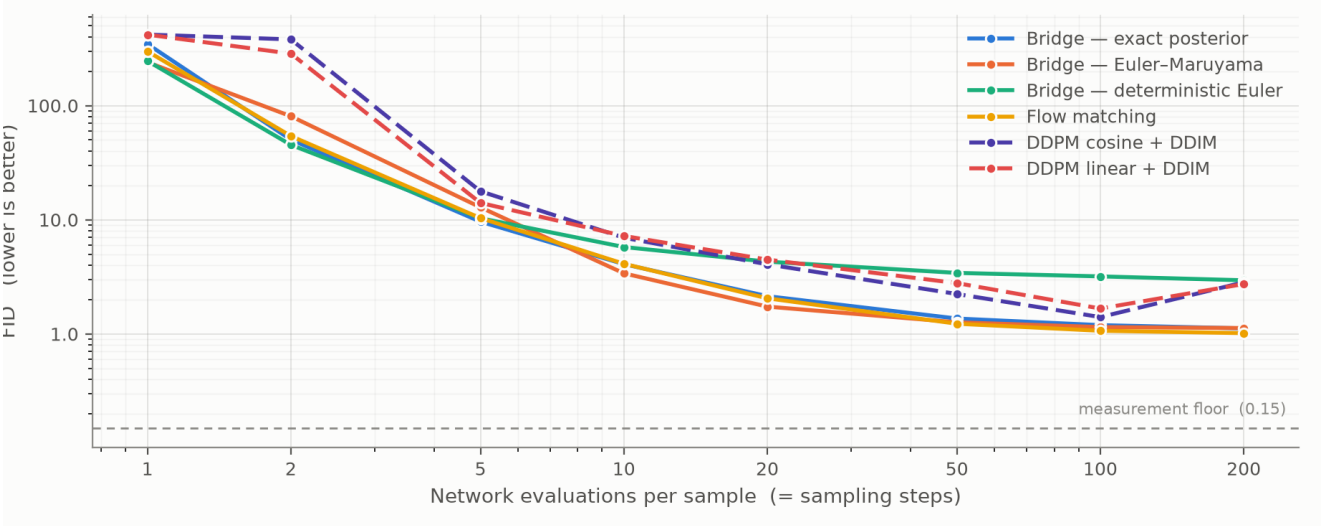
The variance vanishes at $u = 1$: the terminal point is hit exactly, with no $1/(T-t)$ singularity, no boundary offset, and no time-discretisation error. To first order in $(u-s)$ it agrees with (a), so it is the exact form of the same dynamics.

Flow matching is the $\gamma \to 0$, velocity-parameterised member of this family: the interpolant becomes $(1-s)x + sy$ and the targets are related by $v = (\hat{y} - \xi_s)/(1-s)$. A matched-budget comparison therefore isolates exactly the stochastic latent and the parameterisation.

One sampler, three drivers. Scheme (c) needs only the bridge transition kernel, never the martingale $\{M_t\}$ on which the manuscript's Lévy section is gated. Substituting other Lévy marginals gives closed forms with the *same* drift $\xi_s + (\hat{y} - \xi_s)\frac{u-s}{1-s}$ and only the fluctuation changing:

Driver	Exact transition $\xi_s \to \xi_u$	Support	Parameter
Brownian	$\mathcal{N}\big(\text{drift}, \gamma^2 \frac{(u-s)(1-u)}{1-s}\big)$	$\mathbb{R}$	$\gamma^2 = \sigma^2 T$
Gamma	$\xi_s + (\hat{y} - \xi_s)\,\mathrm{Beta}\big(\kappa(u-s), \kappa(1-u)\big)$	$[\xi_s, \hat{y}]$	κ
Poisson	$\xi_s + \mathrm{Bin}\big(\hat{y} - \xi_s, \frac{u-s}{1-s}\big)$	$\{\xi_s, \dots, \hat{y}\} \subset \mathbb{N}_0$	—

The rate parameters (κ's scale, λ) cancel identically; both non-Gaussian rows were verified against the defining construction at 4×10^6 draws per cell.



Sampling efficiency on MNIST at matched budget. FID vs. network evaluations per sample; six schemes sharing one 7.28M UNet, AdamW 2×10^{-4}, EMA 0.9999, batch 128, 40,000 steps and identical scoring code — only the process and its sampler differ. 50,000 samples per point against the full training split; the dashed rule at 0.15 is the floor measured between disjoint real subsets of the same size. Bridge leads at 10–20 evaluations, flow matching from 50. Deterministic Euler (green) is what the original evaluated: it converges to ∼3× the floor excess of the correct samplers. The DDPM curves turn upward after 100 evaluations; checked against an alternative DDIM timestep grid, this is a real consequence of the cosine schedule's near-zero terminal SNR, not an implementation artefact.